AI OPEN
DATA CENTER

Path to the Megawatt Rack: A Grid-Aware Symmetric High-Power UPS for AI Data Centers

Tyler Andrews *VP of Business Dev., Echion Technologies*

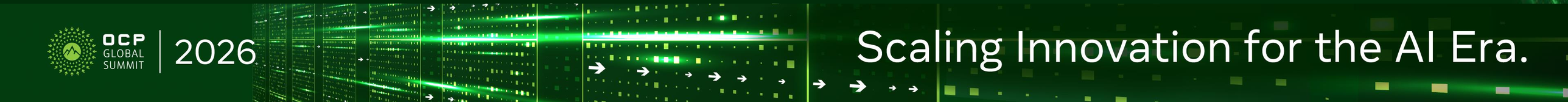
Drew Schaller *Technical Solutions Mngr., Proper Voltage*

OCP BRONZE™ Member

01

The battery's first job was to wait.

It sat fully charged, and once or twice a year it carried the load for a few minutes until the generators came up. Systems have traditionally been sized for power and energy, not for rapid cycling.



Then AI changed the power profile

01

Demand

U.S. data-center electricity, 176 TWh in 2023, is heading to 325 to 580 TWh by 2028, 74 to 132 GW of connected demand (LBNL, 2024). The load is scaling faster, with more induced variability, than the grid was planned to follow.

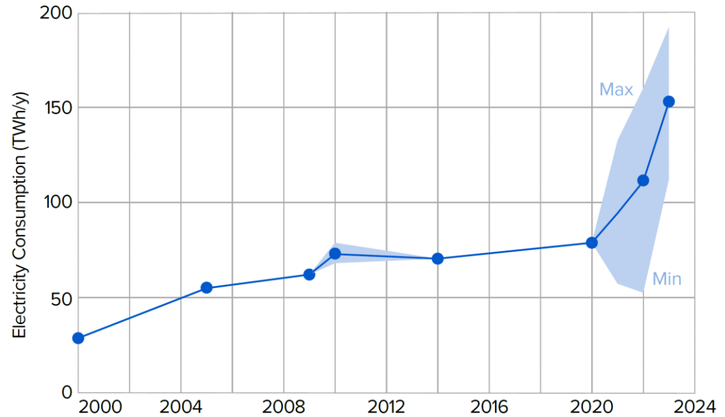
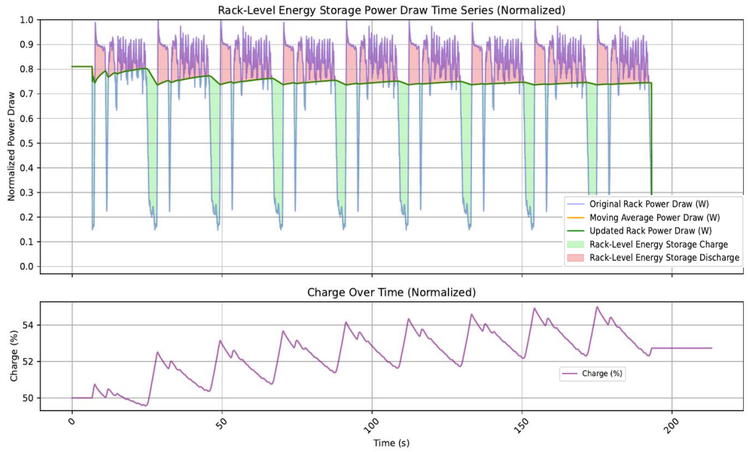


Figure 6. U.S. data center load growth from 2000 to 2023. The graph's light blue area indicates the uncertainty range based on two datasets estimating recent-to-current data center loads [4, 8, 9]

02

Volatility

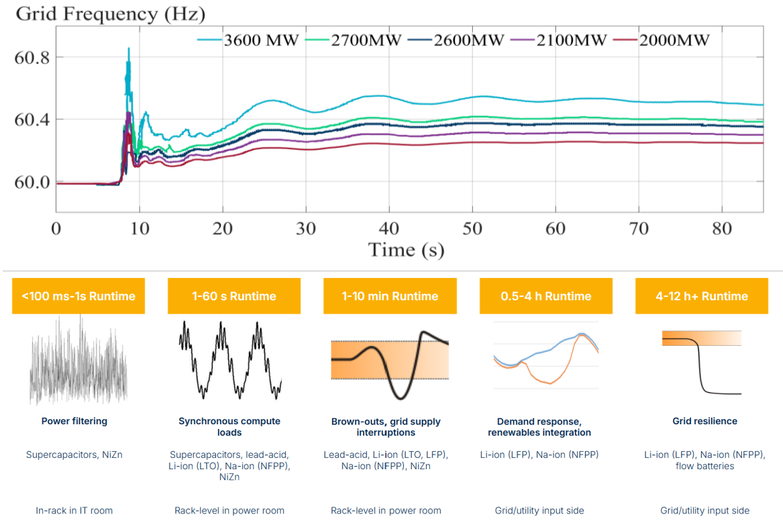
Synchronized GPU clusters swing facility power 30 to 60 percent within milliseconds. Most of the oscillation energy sits at 0.2 to 3 Hz, overlapping turbine and long-line resonant modes; the band operators write limits against is wider, roughly 0.1 to 20 Hz



03

Grid response

The grid is pushing back with ride-through and ramp-back requirements for large loads. Proposed by ERCOT (2025); already required by RTE, Energinet and EirGrid.



OCP LVDC WHITE PAPER V1.0 (MAR 2026) AND 2025 SUMMIT RACK BBU / CBU WORK. RANKING: CHOUKSE ET AL., ARXIV:2508.14318 (2025), TABLE I.

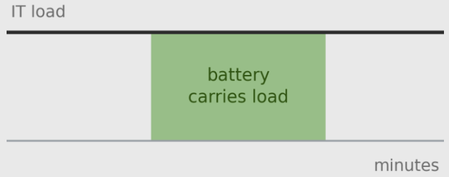
THE SECOND JOB

UPS batteries got a second job, and it scales as rack power increases

JOB 1 · ONE-WAY · RARE

Outage ride through

Carry the critical load at full power until standby generation picks up. Minutes of runtime, once or twice a year, discharge only.

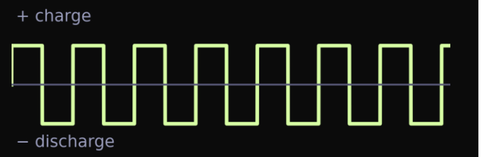


SIZED BY ENERGY

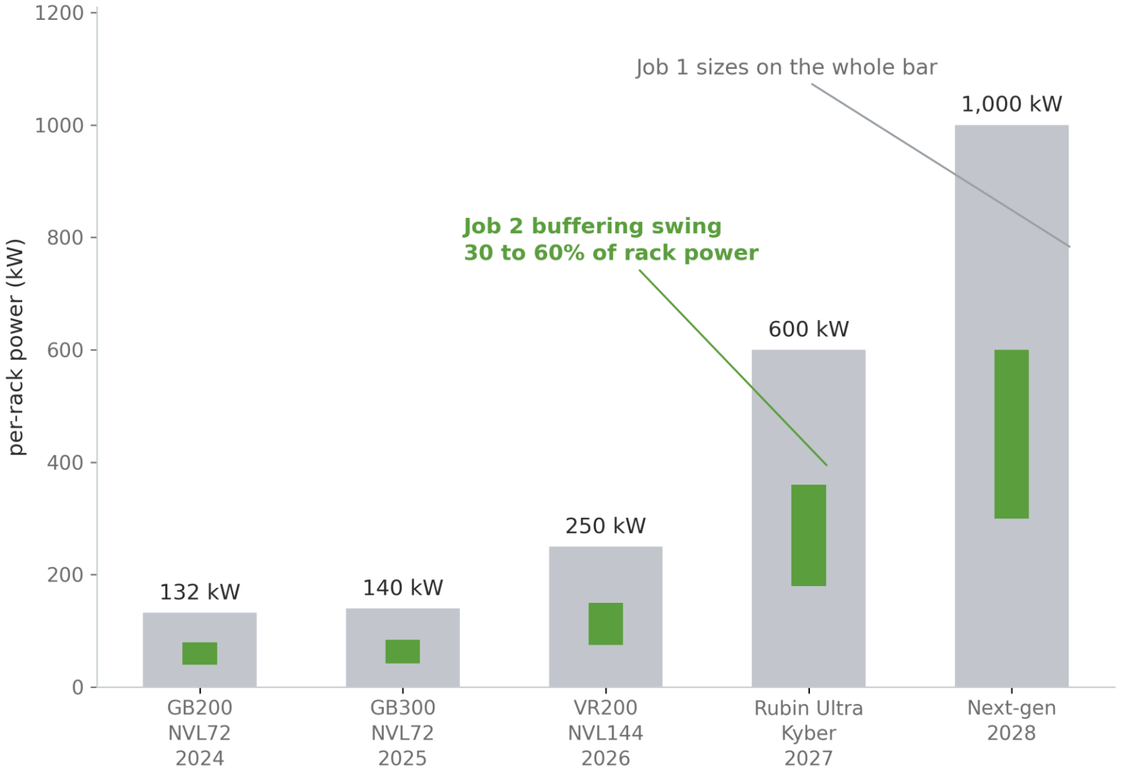
JOB 2 · TWO-WAY · CONTINUOUS

AI load buffering

Charge on every dip, discharge on every spike, many times a minute, for years, at reserve state of charge.



SIZED BY SYMMETRIC POWER AND CYCLE LIFE



Rack power per published roadmaps. Swing band after Choukse, Warrier et al. (Microsoft / OpenAI / NVIDIA), arXiv:2508.14318, 2025.

Why the UPS is being used for both jobs

µs to ms

Transients

Capacitor shelves in the rack

0.1 s to seconds

Rack one way backup

BBU on the rack DC bus

Seconds to minutes

Ride-through and facility buffering

BOTH JOBS · UPS LAYER

Continuously energized, electrically present at every instant

Minutes to hours

Site peaks

Utility-scale BESS containers

Hours +

Extended outage

Backup power generation

AFTER THE OCP LVDC WHITE PAPER V1.0 (MAR 2026) AND 2025 SUMMIT RACK BBU / CBU WORK.

THE GAP AND THE MISSING LINK

What the job demands, and what can meet it

		LEGACY TECHNOLOGY		NEXT GENERATION
		The capacitor	The battery	**The hybrid Battery**
01	**Symmetric power** Energy in equals energy out, at any state of charge	symmetric by nature	charges far slower than it discharges	symmetric by nature
02	**Usable power density** Continuous power at any SOC	high power for very short durations	derated capacity at higher power	continuous high power
03	**Life under duty cycle** Throughput and calendar life	millions of cycles	854 to 2730 cycles at 5C/5C (see slide 8)	50,000 cycles at 5C/5C (see slide 8)
04	**Energy density** Wh/L per cell	5 to 10 Wh/L	300 to 714 Wh/L, depending on chemistry	250 - 300 Wh/L, 30-60x a capacitor

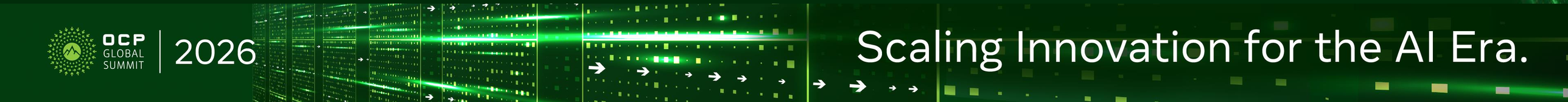
Retention and cycle-life figures from Echion cell characterisation and the BESS simulation-tool cycle-life law; incumbent figures from the published 21700 power-cell datasheet. Floor-area comparison from the 270 MW-class case.

02

Evidence

Cell-level rate data, 1C to 20C

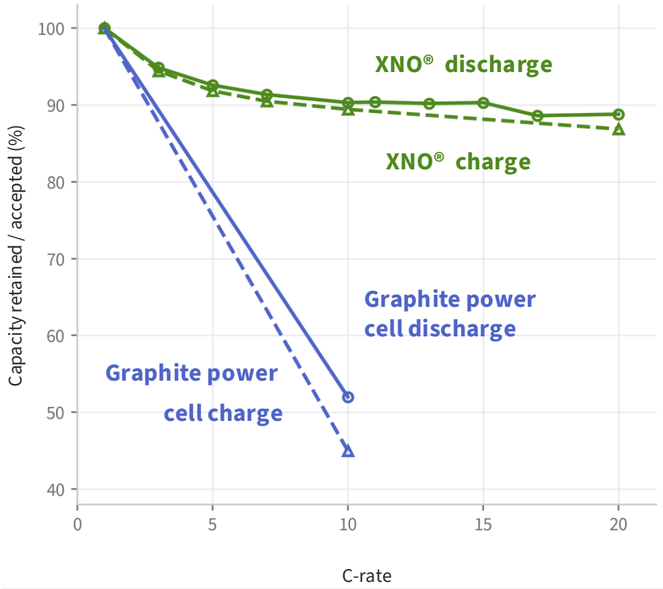
Production consistency, life and abuse



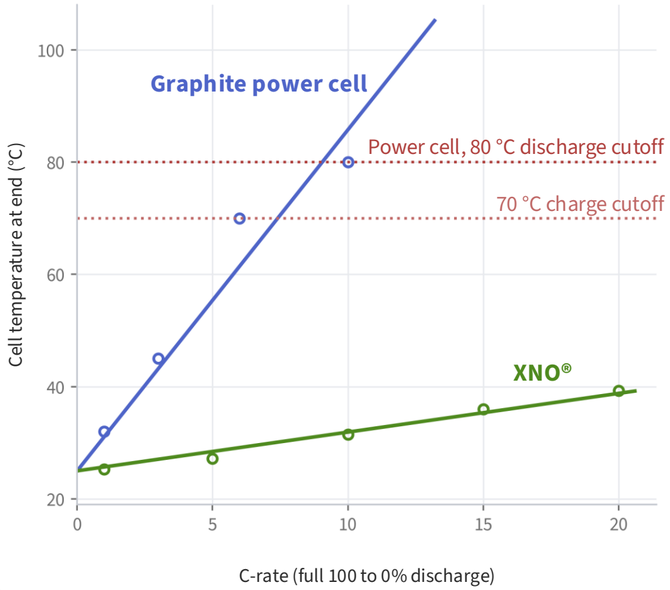
EVIDENCE

Rate, heat and charge acceptance at cell level

Capacity vs rate



Heat under duty



Cycle life at 5C charge and 5C discharge

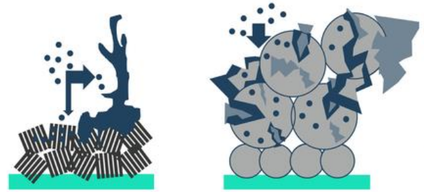
25 °C · 10%–90% SoC (DoD 0.80) · to 80% retention — one square = 1,000 cycles

NMC, high-power 21700
Molicel P50B, graphite
854

LFP, incumbent UPS
1C-class cell, graphite
2,730

XNO®
niobium oxide, ~1.6 V vs Li
50,000

Common Graphite Deg. Factors



Echion BESS simulation-tool cycle-life law, cycles = nameplate × (refDoD/DoD)^k × (refC/C)^m. This is XNO's own rating window, so the model returns the nameplate 50,000 exactly, which is the figure on the cabinet slide. LFP and XNO run on the tool's own parameters; P50B added in the same form (750-cycle anchor at 1C/5C, k 1.70, m 0.155). Both graphite figures are optimistic upper bounds: real graphite plates lithium at 5C.

Symmetric Power: >30C/30C

Low DCIR means thermally efficient cells that can run at high power continuously

Long Life: 50,000 cycles

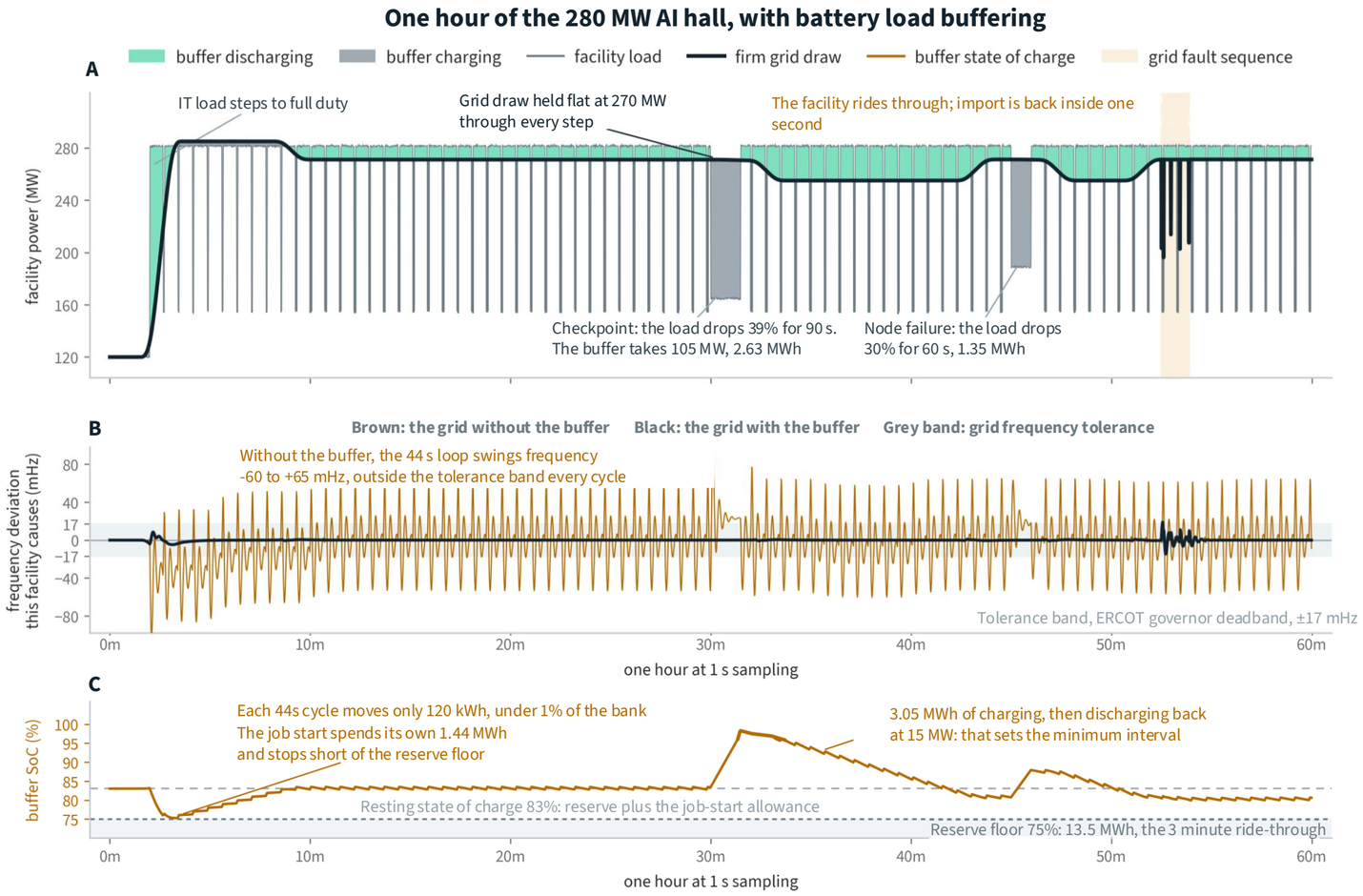
At 10 to 90% SoC, lifetime expectancy is around 50,000 cycles (5C/5C, 25°C)

Safe: EUCAR 3

Nail penetration at full charge, no fire, no explosion, no dendrite formation, and no SEI layer

5.5 AH XNO® CELL MEASURED IN ECHION'S LAB AT 25 °C; LEADING 21700 POWER CELL FROM ITS PUBLISHED DATASHEET. CONSISTENCY, LIFE AND ABUSE DATA: ECHION TECHNOLOGIES, 2026.

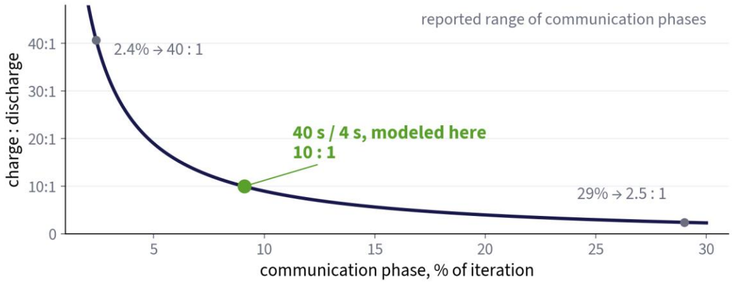
Facility power profile



Recharge is the binding constraint by a factor of 10:1

Peak IT load, compute phase	**280 MW**
Floor, communication phase	**160 MW**
Grid draw, held flat	**270 MW**
Battery Buffer out, 40 s (compute)	**10 MW**
Battery Buffer in, 4 s (communication)	**110 MW**
Energy, each direction	**120 kWh**
Charge to discharge, 40 s out vs 4 s in	**10 : 1**

Charge-to-discharge ratio vs communication phase



As the dot moves right there is less compute and more communication, and charge matters less. Shorter communication phases push the dot left and ratio up. Even at the far right, charge is 2.5 times discharge.

POWER MODELS

Reduce Overbuild based on charge C-rate

The same job over ten years: 1,964 buffering cycles a day, 867 GWh delivered.

CHARGE-RATE CEILING	INSTALLED ENERGY	INREQUIRED CABINETS	REPLACEMENT FREQUENCY	REPLACEMENTS OVER 10 YEARS	COST PER MWh DELIVERED
0.6C **ORV3-class rack BBU** high-power NMC	**183 MWh**	**803 cabinets**	**4 yr**	**2**	**$317**
1C **Incumbent LFP UPS** LFP	**110 MWh**	**860 cabinets**	**6 yr**	**1**	**$25**
6.1C 10C rated **VaultX** XNO®, niobate anode	**18 MWh**	**225 cabinets**	**10 yr**	**0**	**$10**

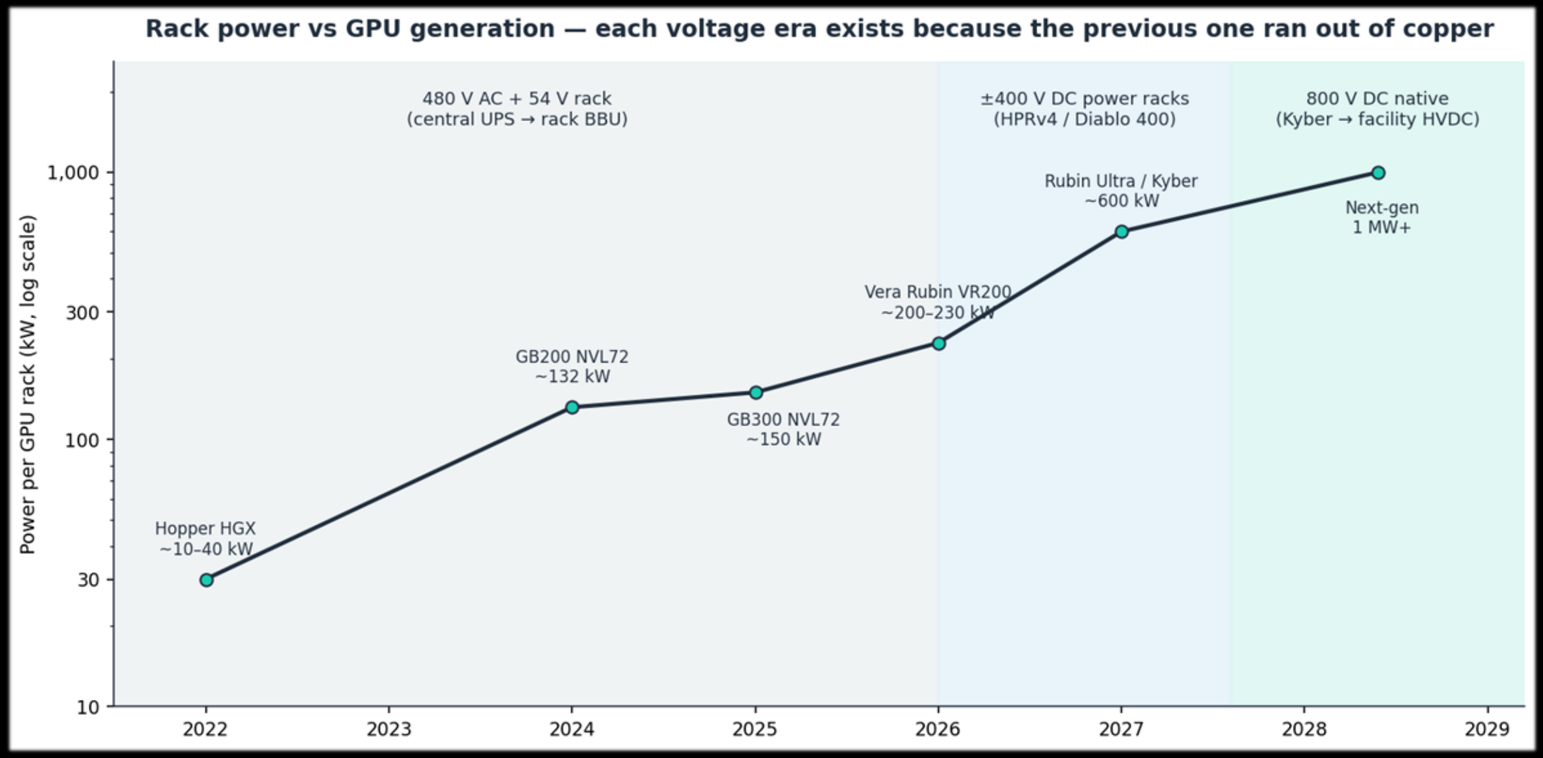
Around 6C symmetric carries both jobs in one bank. Above that, the four-minute ride-through already sets the size, so extra rate buys little.

Service life sourced per row: BBU from the OCP ORV3 BBU module specification, 4 yr. LFP from the BESS simulation at this duty, cycle-limited at 5.6 yr and rounded up to 6. VaultX understated: 8.3 yr on the published 50,000-cycle rating, 13.0 yr from measured cells, shown as 10. Cost is cells only, undiscounted. Cabinet counts use the same 600 x 1200 x 2265 mm enclosure. Installed energy follows from the 110 MW recharge duty divided by the charge-rate ceiling.

03

Deployment

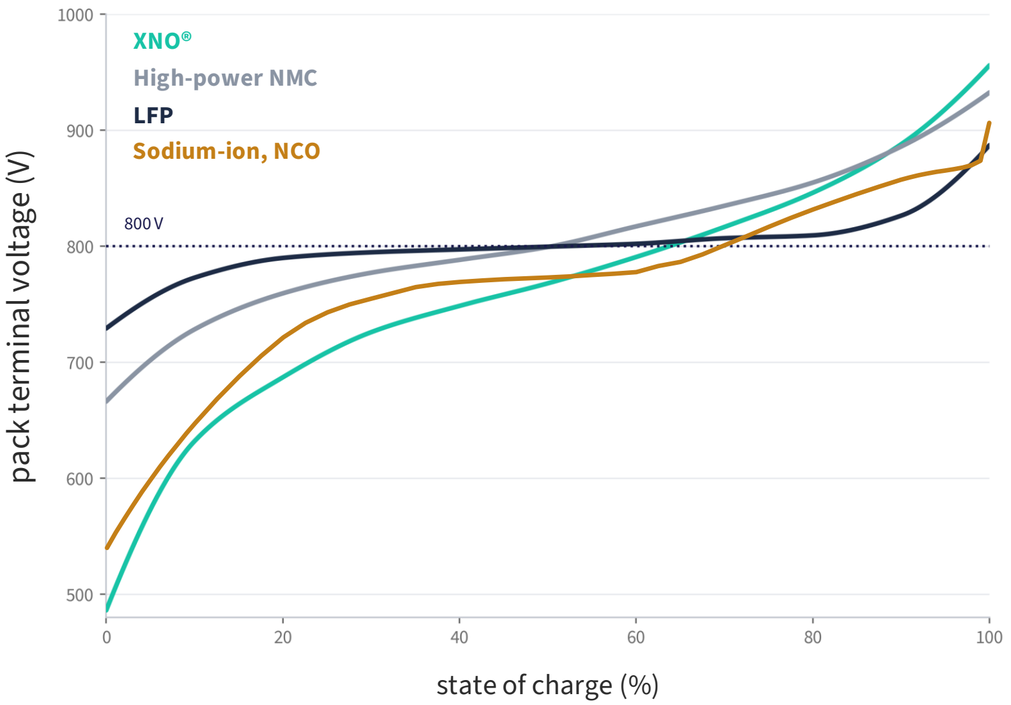
The road to the DC bus



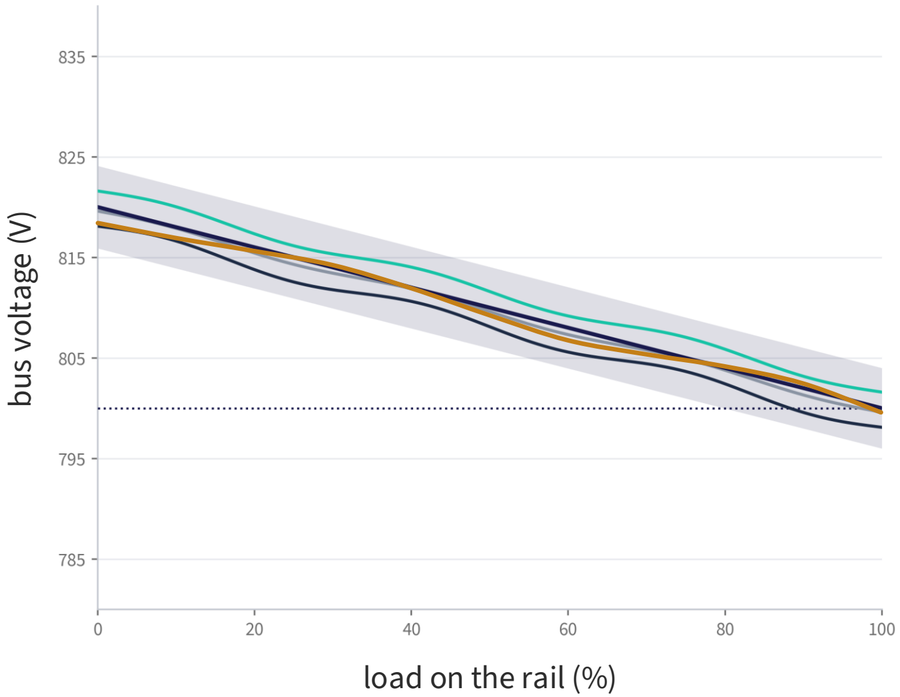
VOLTAGE CONTROL

Any chemistry, on whatever curve the bus specifies

Each chemistry has its own voltage profile



With the battery operating system (bOS)



XNO® curve is the 324S string from the Echion AI data center power architecture primer v3, section 7. LFP and NMC curves are ILLUSTRATIVE shapes at published cell windows, series-counted to sit near 800 V at mid-charge.
they show that no chemistry is flat, they do not characterize those cells. Rail command and the ±0.5% static band from the OCP Diablo 400 base specification v0.7.0, Table 4. Regulation error drawn indicatively, not from a Proper Voltage datasheet. Sodium-ion: measured 70 Ah cell, 270S.

A pack that presents whatever voltage the bus asks for, whatever chemistry sits behind it.

DEPLOYMENT

Cabinets for 480 & 800VDC architectures



1.6 MW Peak / 1 MW Continuous

Prototype build underway

Form factor	48U cabinet
Voltage	480 & 800 VDC Nom.
Cycle life	50,000+ at 5C/5C, 10 to 90%
Energy	80 kWh
Build	Config up to 12 packs, active cooled
Safety path	UL 9540, 9540A, 1973, UN 38.3

STEP 1 · TODAY

Unregulated in the UPS

Battery system drops into the existing UPS as native 480VDC storage, with no added power electronics. It carries both duties, and nothing else in the facility changes.

STEP 2 · AS 800 VDC ARRIVES

A regulated battery at the DC bus

As distribution goes to 800 VDC, the battery moves next to the racks, behind power electronics that regulate its voltage onto the bus. DC/DC conversion moves into the servers.

PRELIMINARY, SPECIFICATIONS SUBJECT TO CHANGE.

THE ASK

Call to Action: Energy - Open Compute Project

OPERATORS

Pilot grid-aware UPS behavior with us

Test ramp-rate and ride-through behavior at the UPS layer against your interconnection requirements, and tell us what the specification should say.

UPS OEMS

Co-validate the cabinet, or the layer

Joint system validation in 2027, results co-authored. Or keep your own platform and we adapt the control layer to it.

OCP COMMUNITY

Specify the duty cycle for current and future AIDCs

Study the storage requirements for symmetric-duty UPS (480V) and Sidecar (400V & 800V) for DCF Power Distribution.

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Thank You!

Open discussion

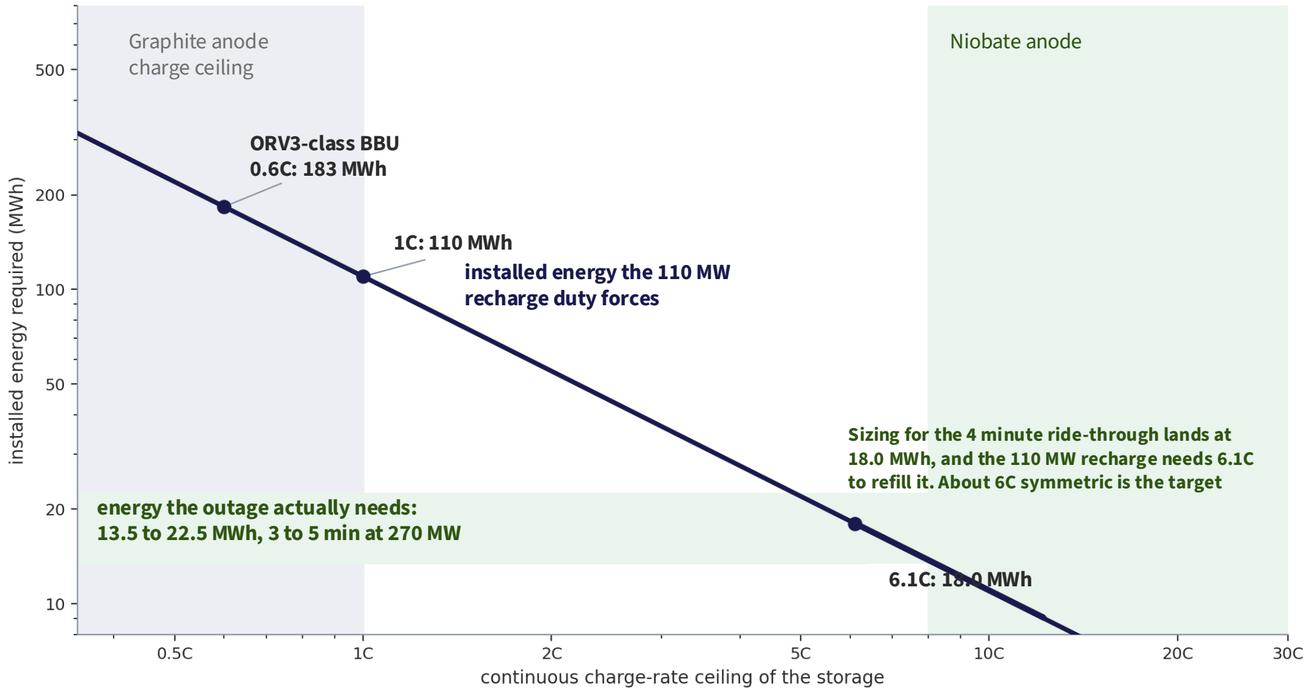
Tyler Andrews, Echion Technologies · Drew Schaller, Proper Voltage

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POWER MODELS

Overbuild also sets how often you buy it again



The same job, over ten years

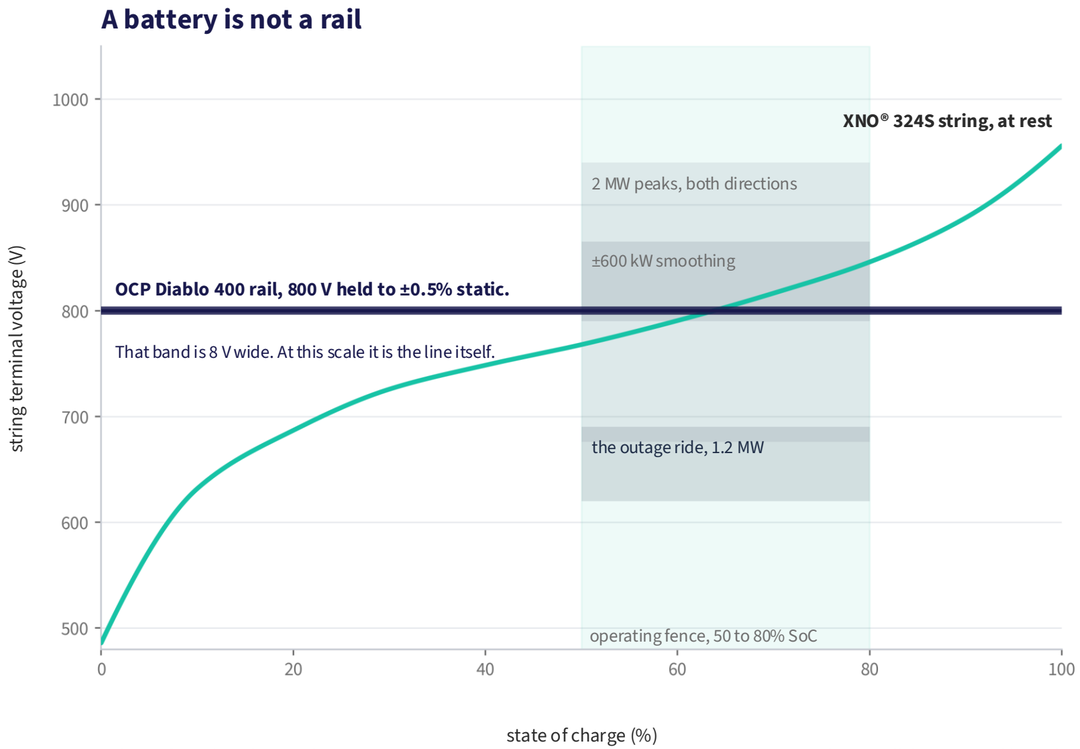
1,964 buffering cycles a day, 867 GWh delivered over 10 years.
Each service life below comes from its own source, not from one model.

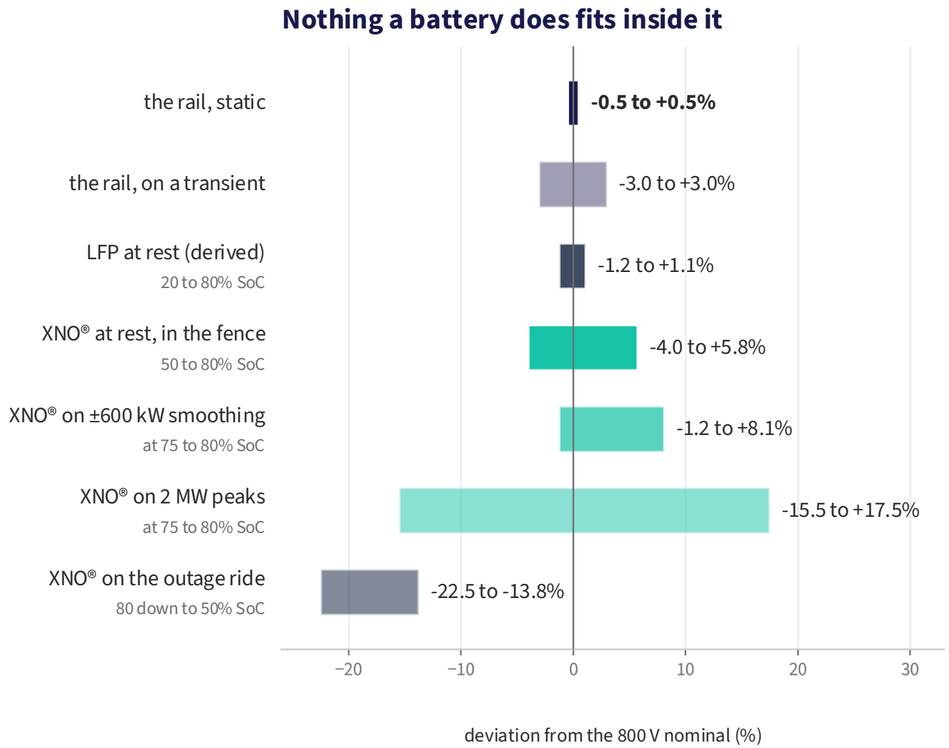
	installed	replaced every	builds in 10 years	cells over 10 years	per MWh delivered
ORV3-class rack BBU					
0.6C ceiling · high-power NMC · 803 cabinets	**183.3 MWh**	**4 yr**	**3**	**$275 M**	**$317**
OCP ORV3 BBU specification, 4 year service life					
Incumbent LFP UPS					
1C ceiling · LFP · 860 cabinets	**110.0 MWh**	**6 yr**	**2**	**$22 M**	**$25**
simulation at this duty retires it on cycles at 5.6 yr, rounded up					
VaultX					
6.1C used, 10C rated · XNO · 225 cabinets	**18.0 MWh**	**10 yr**	**1**	**$9 M**	**$10**
one build, and it is still running at year 10					

Service life sourced per row: BBU from the OCP ORV3 BBU module specification; LFP from the BESS simulation at this duty, cycle-limited at 5.6 yr and rounded up so the incumbent gets its best case; VaultX understated against 8.3 yr on the published 50,000-cycle rating and 13.0 yr from measured cells. Cells only, undiscounted. One square is one cabinet.

BACKUP

Why a regulated DC bus cannot take a battery directly



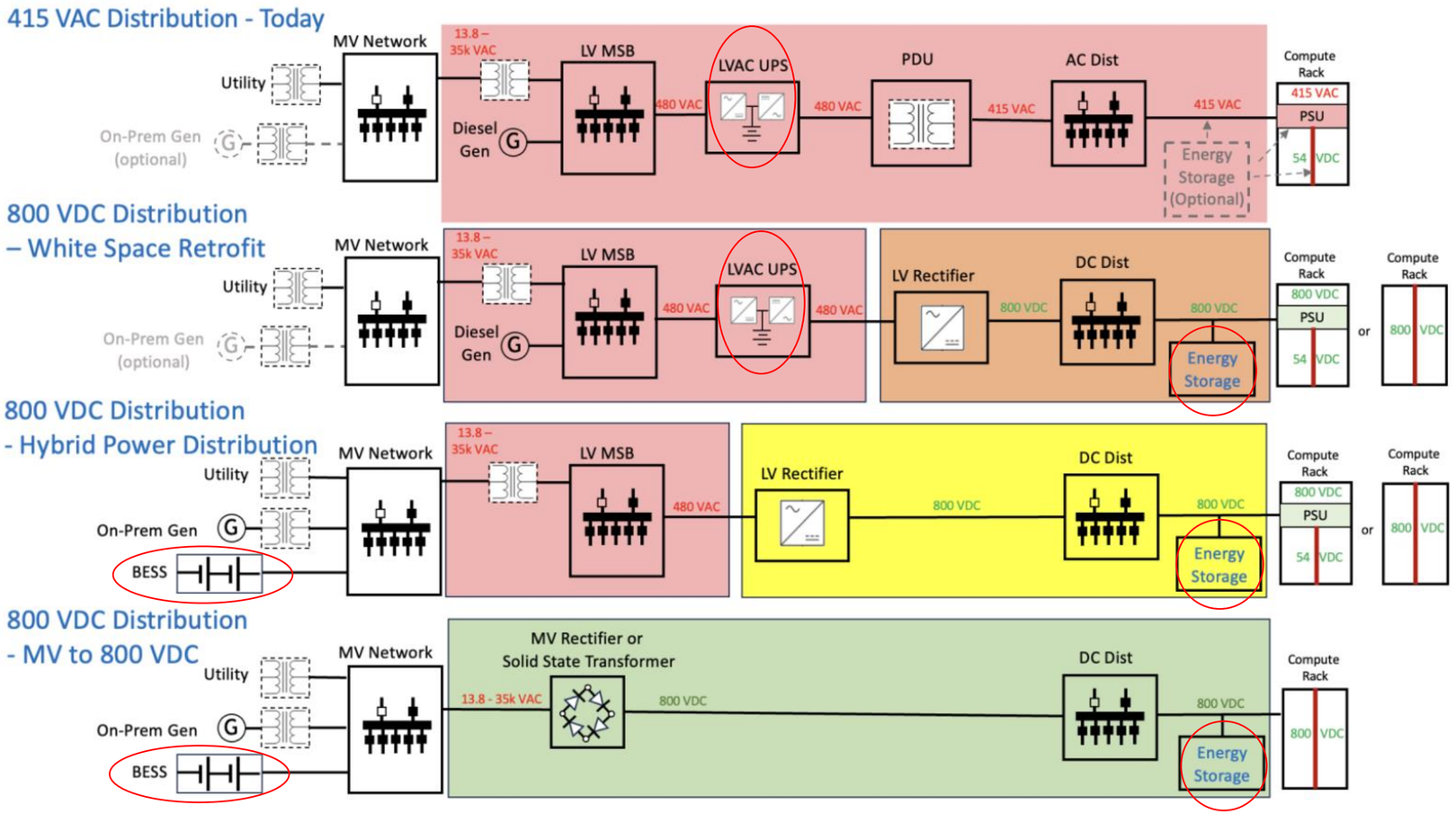


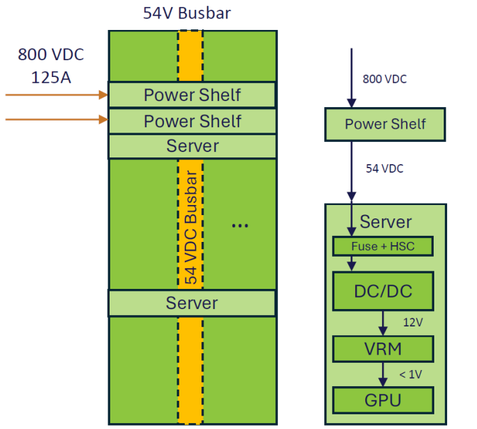
Voltages from the Echion AI data centre power architecture primer v3, section 7, on a 324S string at four parallel strings. Rail limits from the OCP Diablo 400 base specification v0.7.0, Table 4.

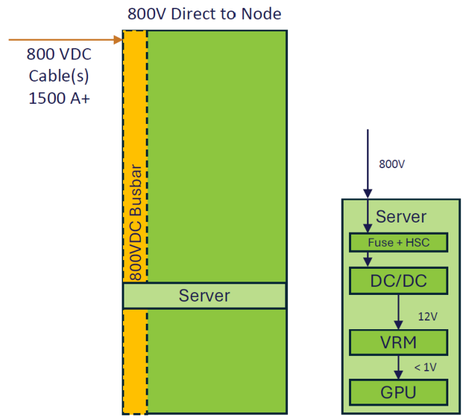
The LFP row is derived here: a 243S string on published LFP open circuit voltage. The outage row's 80 to 50 per cent span is derived too, from the sag being a consistent 150 V at 1.2 MW across the fence.

A DC-DC stage is not an option on this bus. It is the condition of entry, and it applies to every chemistry, not just ours.

Progression of power distribution in AI







Source: 800 VDC MGX accelerated computing rack energy storage for improved GPU power utilization.

BACKUP

Grid codes and references

GRID REQUIREMENTS

ERCOT (LLWG, 2025): ~2,600 MW of instantaneous load loss can push frequency to ~60.4 Hz with RoCoF above 5 Hz/s; voltage ride-through proposed for large electronic loads.

RTE, Energinet and EirGrid: large loads must ride through faults and ramp back, roughly 90 to 100% of pre-fault power within about one second in some regimes.

Emerging utility specifications: ramp-rate limits in MW/s, permitted dynamic power range, and frequency-domain caps across roughly 0.1 to 20 Hz.

REFERENCES

Choukse, Warrier et al. (Microsoft, OpenAI, NVIDIA), Power Stabilization for AI Training Datacenters, arXiv:2508.14318, 2025.

Shehabi et al., 2024 United States Data Center Energy Usage Report, LBNL-2001637, 2024.

Open Compute Project, LVDC white paper v1.0, March 2026; Mt. Diablo power architecture specification.

NERC, Level 3 Essential Actions Alert on computational load loss events, May 2026.

Echion Technologies, XNO® for AI data centres, 2026; cell specification and characterisation data, 2026.